# A kinetic-impact deflection model

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Estimate a velocity change and explain the value of lead time.

**SUCCESS CRITERIA - I can...**

1. State the simplified collision model.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “deflection” and “lead time” to explain your model or result. Sentence starters are optional.

**deflection**: a change in the direction or path of motion

**lead time**: time available before the modeled event

**impulse**: net force acting over time; change in momentum

## 01 Bring back an earlier idea

A 0.3 kg projectile at 50 m/s sticks to a stationary 2.7 kg cart.
**Find** common speed and the fraction of initial kinetic energy remaining.

**Word help:** initial = at the beginning of the interval; subscript i

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × lead time. Toy model: target mass is much larger; straight-line shift comparison. Ejecta can also transfer momentum in a real impact.

**Worked reasoning:** In a fictional model, a 2 kg projectile at 30 m/s sticks to a stationary 10 kg target. Final speed is 60/12=5 m/s. The system includes both objects.

**Your turn: Draw** the before-and-after system boundary and momentum arrows.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

A tiny sideways velocity change acts for a short time or a long time. Which produces more sideways shift in the straight-line model?
**Explain**.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** ① ② ③

## 04 Solve and justify

Hypothetical impactor: 600 kg at 6000 m/s; target $5\times10^9$ kg initially at rest in the target frame. Ignore ejecta. EstimateΔv and straight-line shift after $10^6$ s.

**Word help:** at rest = not moving in the chosen reference frame

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Analyze the reasoning

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

An asteroid's crossing window is days 16-20; Earth's is 18-22.
**Find** the overlap and explain why this is not proof of a collision.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** 1 2 3

## 07 Exit check

A toy target needs a 10 km shift over $2\times10^{6}$ s.
**Find** required$\Delta v$. If target mass $10^{9}$ kg, estimate impulse.

**Word help:** estimate = find an approximate value

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

A 1000 kg hypothetical target needs a transverse velocity change 0.02 m/s.
**Estimate** required impulse. If delivered uniformly for 0.1 s, find average force and displacement from the velocity change after $10^{5}$ s in a force-free toy model.

**Word help:** estimate = find an approximate value

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** 1 2 3